

Preparation of Cas9 Protein

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Materials

TY media	1L	
Tryptone	10	g
Yeast extract	6	g
NaCl	10	g
Dissolve in 400mL		
Rinse flasks thoroughly with MQ water		
Add 600mL MQ water, media		
Autoclave (45', no dry time)		

Buffers

Lysis – 1L, pH7.5

20mM HEPES	20	mL of 1M
1M KCl	74.55	g
10% glycerol	100	mL
5mM TCEP	1.43	g
10mM imidazole	680	mg

Elution – 500mL, pH7.5

20mM HEPES	10	mL of 1M
0.1M KCl	50	mL of 1M
10% glycerol	50	mL
5mM TCEP	717	mg
300mM imidazole	10.2	g

Ion Exchange buffer A – 1L, pH7.5 (low salt)

20mM HEPES	20	mL of 1M
300mM KCl	300	mL of 1M
10% glycerol	100	mL
1mM TCEP	287	mg

Ion Exchange buffer B – 1L, pH7.5 (high salt)

20mM HEPES	20	mL of 1M
1M KCl	74.55	g
10% glycerol	100	mL
1mM TCEP	287	mg

Gel Filtration – 1L, pH7.5

20mM HEPES	20	mL of 1M
150mM KCl	150	mL of 1M
10% glycerol	100	mL
1mM TCEP	287	mg

Expression of Cas9 in Rosetta2 *E. coli*

Transformation

1. Mix 1uL of 10ng/uL DNA + 10uL Rosetta 2 cells (pRAR-rare codons).
2. Heat shock: 10' ice, 45" 42°C, 5' ice.
3. Recover: Add 300ul LB, 50' 37°C on shaker.
4. Spin down (6000xg, 1'), and remove 150ul LB.
5. Plate 100ul, 30ul on 2 AMP plates.

Starter culture

1. Pick 6 colonies to add to 20mL LB + Amp (Cf=100ug/mL) starter culture.
2. Grow overnight.
Next day: Store XL1 blue 20% glycerol stocks in -80°C.

Expression

1. Add AMP to the flasks (Cf=100ug/mL).
2. Add 20mL starter culture in the morning,
3. Grow to OD600=0.6 (150rpm, ~4hrs, 37°C)
4. Set shaker temp to 16°C.
5. *Save 1mL for SDS-PAGE: **uninduced**.
6. Add 0.5mM IPTG.
7. Grow overnight (150rpm, at least 16hrs, 16°C).

Next day:

8. Harvest cells before lunch: Spin down (4000rpm, 20', 4°C). ***Keep the cells cold from here.**
9. Completely re-suspend pellets in 20mL Lysis Buffer + 1mM PMSF per bottle.
10. Use additional 20mL Lysis Buffer to wash all bottles.
11. Transfer to 50mL-tubes.
12. *Save 20uL for SDS-PAGE: **cell pellet**
13. Freeze the cells at -80°C.

Purification of Cas9

Sonication

1. Thaw cells in water.
2. Push beaker with a stir bar to the bottom of ice bucket.
3. Add ~50mL Lysis Buffer and 1 protease inhibitor tablet/50mL cell. Dissolve the tablets.
4. Transfer the cells to the beaker.
5. Sonicate while stirring: 8' total, 10" on, 10" off, level 4. Lysate should turn fluid and thin. Repeat 8min cycle, if necessary.
6. Transfer to oakridge tubes (30mL/tube). *Balance within +/- 0.01g.
7. Pellet cell debris (18,000rpm, 30', 4°C). Prepare HisTrap purification.
8. *Save 20uL supernatant for SDS-PAGE: **cell lysate**

HisTrap

CV=5mL, FR=2mL/min

Buffer A: Lysis buffer, Buffer B: Elution buffer

1. Filter sample.
2. Set-up Akta:
 - a. Turn on UV lamp.
 - b. Wash Akta with water: pump wash, and flow 50%A, 50%B.
 - c. Pump wash.
 - d. Flow 50%A, 50%B.
 - e. Flow 100%A, until steady baseline. Set baseline to zero.
3. Apply sample with pump, in the coldroom:
 - a. Fill tubing with water, connect column.
 - b. Wash: 5CV water.
 - c. Equilibrate: 5CV lysis buffer.
 - d. Apply sample to the column. Collect flow-through: Ni-sample loading. Re-apply flow-through to the column, if necessary.
4. Purification:

In the coldroom:

 - a. Wash: 15CV lysis buffer. Collect flow-through: Ni-wash1. Collect flow-through: **Ni-wash1**. *Save 20uL for SDS-PAGE.
 - b. Wash2: 3CV lysis buffer + 0.1% Triton X. Collect flow-through: **Ni-wash2**. *Save 20uL for SDS-PAGE.

Using Akta:

 - c. Attach column to Akta (FR=0.9mL/min).
 - d. Wash: 5CV lysis buffer, until steady UV baseline.
 - e. Elute: 25CV elution buffer, 100% gradient.
 - f. Collect fractions (Fraction size=2.5mL).
5. Clean-up:
 - a. Wash: 5CV water
 - b. Store: 5CV 20% ethanol
 - c. Detach column.
 - d. Wash Akta with water.
6. Pool appropriate fractions. Rinse collection tubes with elution buffer.
*Save 10uL for SDS-PAGE: **Ni-elution**

TEV

1. Concentrate with 30K Amicon to ~3mL volume. Collect flow-through: Ni-Amicon FT.
2. Add ~1.25mg TEV/3mL sample, and mix with pipet.
3. Incubate on shaker, at 4°C overnight.

Next day

4. *Save 10uL for SDS-PAGE: **postTEV**
5. Add 25mL of buffer A to sample.

Heparin

CV=5mL, FR=1mL/min

Buffer A: IEX low salt buffer, Buffer B: IEX high salt buffer

1. Set-up Akta (FR=2mL/min).
2. Apply sample with pump, in the coldroom:
 - a. Equilibrate: 10CV buffer A.
 - b. Apply sample to the column. Collect flow-through: Hep-sample loading.
3. Purification:
 - a. Attach column to Akta (FR=0.9mL/min).
 - b. Wash: 15CV buffer A, until steady baseline.
Collect flow-through: **Hep-wash**. *Save 20uL for SDS-PAGE.
 - c. Elute: 20CV buffer B, 100% gradient.
 - d. Collect fractions (Fraction size=2mL).
4. Clean-up (FR=2mL/min):
 - a. Wash: buffer B, until steady conductivity baseline for over 30mL.
 - b. Store: 5CV 20% ethanol.
 - c. Detach column.
 - d. Wash Akta with water.
5. Pool appropriate fractions. Rinse collection tubes with buffer B.
*Save 10uL for SDS-PAGE: **Hep-elution**
6. Concentrate sample to ~1mL. Collect flow-through: Hep-Amicon FT.
7. Leave concentrator on ice, at 4°C overnight.

MBPTrap/Superdex 200

CV=120mL, FR=0.5-1mL/min

Buffer A: Gel filtration buffer

Previous day:

1. Set-up Akta (FR=2mL/min):
2. Prepare columns:
 - a. MBPTrap wash: 5CV water, 5CV gel filtration buffer (FR=2mL/min), in the coldroom.
 - b. Attach MBPTrap and Superdex200 to Akta.
 - c. Equilibrate overnight: 2CV gel filtration buffer (FR=0.3mL/min).
*Set end timer volume=240mL, max pressure alarm=0.25MPa.

Next day:

3. Concentrate sample to ~500uL. Rinse concentrator with 500uL gel filtration buffer.
4. Spin down sample (11000rpm, 4', 4°C). Transfer supernatant to a new 1.5mL-tube.
5. Inject sample:
 - a. Clean injection port, sample loop with water, gel filtration buffer
 - b. Draw up sample with 1mL-syringe, and replace the syringe on the injection port.
 - c. [LOAD] Inject sample to the loop with syringe.
6. Purification
 - a. [INJECT] Apply sample to the column (~3mL).
 - b. [LOAD] Flow ~1CV gel filtration buffer. Cas9 elutes at 65-85mL.
 - c. Collect fractions (Fraction size=2mL).
7. Clean-up (FR=2mL/min):

- a. Detach MBPTrap.
 - b. Wash MBPTrap: 10CV 10mM maltose + gel filtration buffer, in the coldroom.
 - c. Store MBPTrap: 5CV 20% ethanol.
 - d. Wash Superdex200: 2CV water (Long-term storage in 20% ethanol).
 - e. Detach Superdex200.
 - f. Wash Akta with water.
8. Pool appropriate fractions. Rinse collection tubes. *Save 5uL for SDS-PAGE: **Cas9**
9. Concentrate to desired concentration (~40uM).

*Cas9	MW=158,441.4g/mol Molar absorption coefficient=120,575L/mol•cm
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10. Aliquot 20uL each into 1.5mL-tubes.
11. Snap-freeze in LN2, and store at -80°C.